

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 July 2026
Team ID	
Project Name	SHOPEZ: E-commerce Application
Maximum Marks	4 Marks

Technical Architecture:

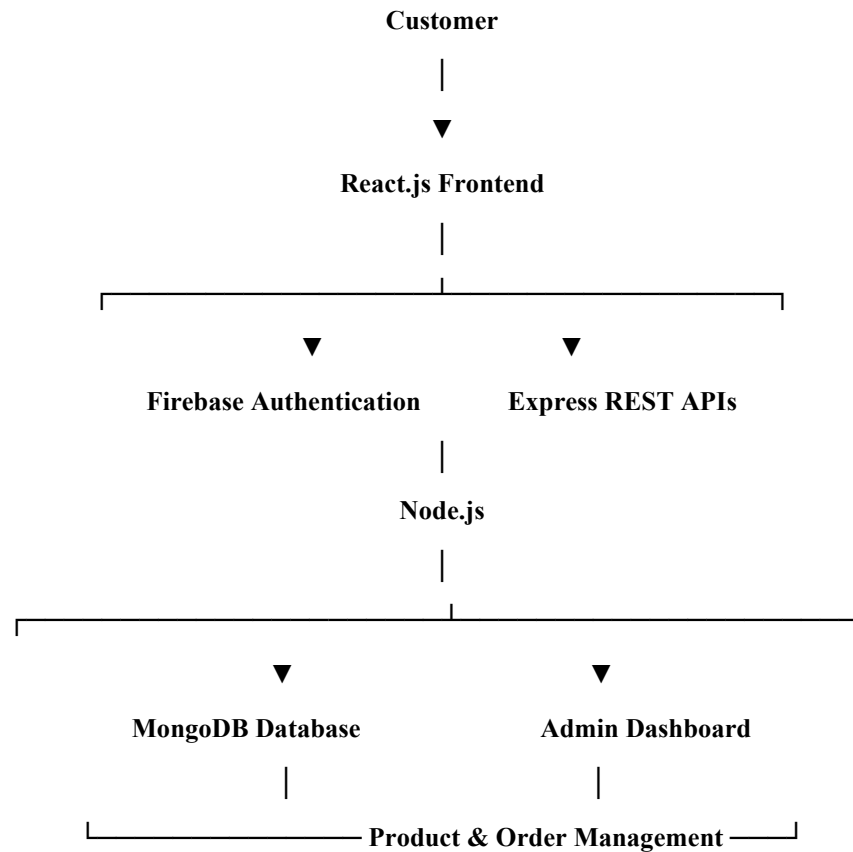


Table-1: Components & Technologies:

S. No	Component	Description	Technology
1	User Interface	Responsive web interface for customers and administrators	React.js, HTML5, CSS3, JavaScript
2	Application Logic-1	User authentication and session management	Firebase Authentication
3	Application Logic-2	Product, cart, checkout and order processing	Node.js, Express.js
4	Application Logic-3	Admin product management (CRUD operations)	Express.js REST APIs
5	Database	Stores product, order and user-related data	MongoDB with Mongoose
6	Cloud Database	Cloud-hosted NoSQL database	MongoDB Atlas
7	File Storage	Stores product images and application assets	Local Storage / Public Assets
8	External API-1	Firebase authentication services	Firebase Authentication API
9	External API-2	HTTP communication between frontend and backend	Axios
10	Machine Learning Model	Not applicable in current version	N/A
11	Infrastructure (Server / Cloud)	Frontend and backend deployment	Frontend: Vercel/NetlifyBackend: Render/Railway/Node Server

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source frameworks used to build the application	React.js, Node.js, Express.js, MongoDB
2	Security Implementations	User authentication, protected admin routes and secure API access	Firebase Authentication, Protected Routes
3	Scalable Architecture	Three-tier architecture separating frontend, backend and database	MERN Stack
4	Availability	Application accessible through cloud deployment with reliable services	Netlify/Vercel, Render, MongoDB Atlas
5	Performance	Fast page rendering, REST APIs, optimized database queries and efficient state management	React.js, Axios, Context API, MongoDB